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Cohort : Jensen Huang

Roll No : 150096724023

Subject : AWS

Problem Statement No : 90

Problem Statement Title : 90. eSportArena Competitive Gaming Cloud

Sem : 4 **, Sprint :** 02

Industry: Esports & Competitive Gaming

Github Repo Link :

Documentation Link :

1. Abstract

eSportArena is a cloud-based gaming platform infrastructure deployed on Amazon Web Services (AWS). The project demonstrates the implementation of core AWS services including VPC, Public Subnets, Security Groups, EC2 Instance, Ubuntu Server, and Nginx Web Server.

The objective is to create a scalable, secure, and highly available cloud environment capable of hosting web applications. The website is successfully deployed on an EC2 instance and accessed through a public IP address.

2. Objectives

- Deploy a website on AWS Cloud.
- Configure networking using VPC and Subnets.
- Launch and manage an EC2 instance.
- Install Ubuntu and Nginx Web Server.
- Understand cloud infrastructure deployment.
- Demonstrate secure and scalable cloud architecture.

3. Technologies Used

Technology	Purpose
AWS EC2	Virtual Server
AWS VPC	Network Isolation
Public subnet	Internet Access
Security Group	Firewall Rules
Ubuntu 24.04	Operating System
Nginx	Web Server
Draw.io	Architecture Diagram
HTML/CSS	Landing page

4. Folder Structure

eSportArena-AWS-Project/

```
│ ├── README.md
├── Architecture.md
├── Project_Report.docx
├── Project_Presentation.pptx
├── website/
│   ├── index.html
│   └── styles.css   (agar CSS alag file me hai)
│ ├── diagrams/
│   ├── aws_architecture.drawio
│   ├── aws_architecture.png
│   └── aws_architecture.pdf
│ ├── screenshots/
│   ├── 01_vpc_created.png
```

```
│   ├── 02_public_subnets.png
│   ├── 03_security_group.png
│   ├── 04_ec2_running.png
│   ├── 05_ubuntu_terminal.png
│   ├── 06_nginx_installed.png
│   ├── 07_website_live.png
│   └── 08_architecture_diagram.png
│ ├── deployment/
│   ├── deployment_steps.md
│   ├── nginx_commands.txt
│   └── aws_configuration.txt│
└── docs/
    ├── Abstract.md
    ├── Objectives.md
    ├── Results.md
    ├── Future_Scope.md   └── Conclusion.md
```

4. Architecture Overview

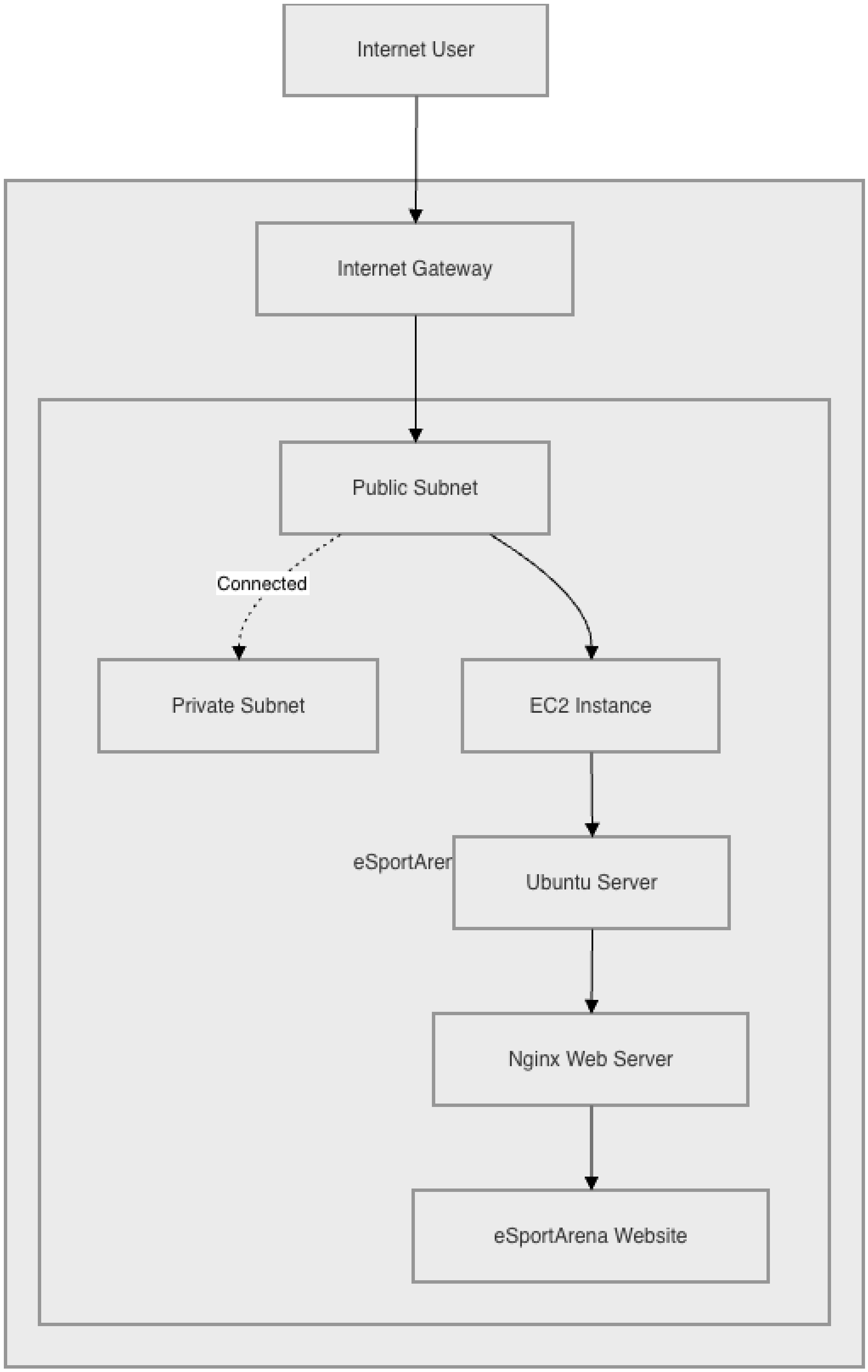


Figure 1: AWS Cloud Architecture of eSportArena Platform

5. Implementation Steps

Step 1: Create VPC

Created a Virtual Private Cloud to isolate network resources.

Step 2: Create Public Subnets

Created two public subnets for resource deployment.

Step 3: Configure Security Groups

Allowed SSH (Port 22) and HTTP (Port 80) traffic.

Step 4: Launch EC2 Instance

Created an Ubuntu-based EC2 instance.

Step 5: Install Nginx

Installed and enabled Nginx web server.

Step 6: Deploy Website

Uploaded a custom HTML page and hosted it using Nginx.

Step 7: Test Deployment

Verified successful deployment using the public IP address.

Step 8: Configure MySQL Database

Installed and configured MySQL Database Server on the EC2 instance.
Verified database service status and connectivity.

Step 9: Linux User Management

Created a new Linux user and configured user access permissions for system administration tasks.

Step 10: Permission Management

Assigned appropriate file and directory permissions to ensure secure access control.

Step 11: Configure Cron Jobs

Created and tested cron jobs for automated task scheduling and system administration activities.

Step 12: Install Docker

Installed Docker Engine on the Ubuntu EC2 instance and verified successful installation.

Step 13: Deploy Docker Container

Created and executed a Docker container to demonstrate containerized application deployment.

Step 14: Create Shell Script Automation

Developed and executed shell scripts to automate administrative and deployment-related tasks.

Step 15: Create IAM User

Created an IAM user with appropriate permissions to demonstrate AWS Identity and Access Management concepts.

Step 16: Create Amazon S3 Bucket

Created and configured an Amazon S3 bucket for cloud storage demonstration.

Step 17: Configure CloudWatch Monitoring

Accessed CloudWatch Metrics and monitored EC2 instance performance and resource utilization.

Step 18: Create AWS Architecture Diagram

Designed the complete AWS architecture diagram using Draw.io, illustrating VPC, Subnets, Security Groups, EC2 Instance, Ubuntu Server, Nginx Web Server, and Website Deployment.

Step 19: Prepare Documentation

Created README.md, Architecture.md, deployment documentation, architecture diagrams, screenshots, and final project report.

Step 20: Upload Project to GitHub

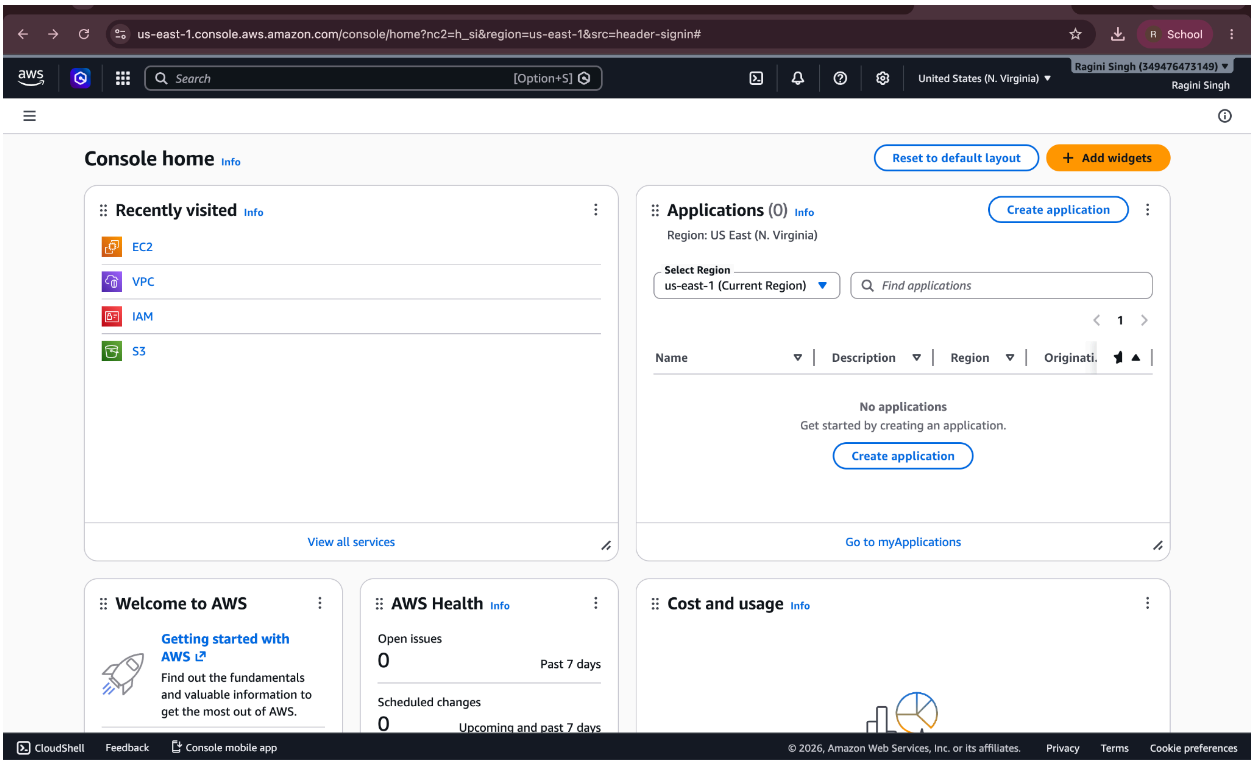
Created a GitHub repository and uploaded source code, architecture diagrams, documentation files, and deployment screenshots.

Step 21: Final Verification

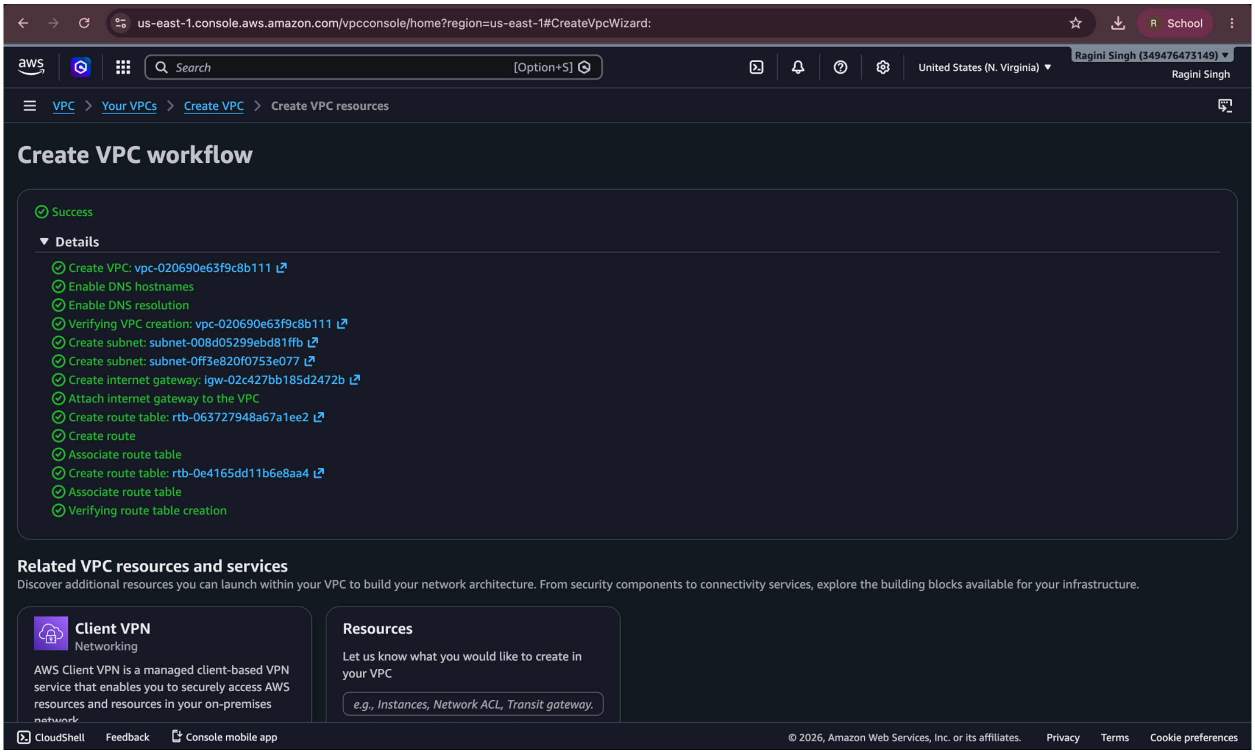
Verified AWS resources, website deployment, monitoring configuration, documentation, and repository structure before project submission.

6. Screenshots

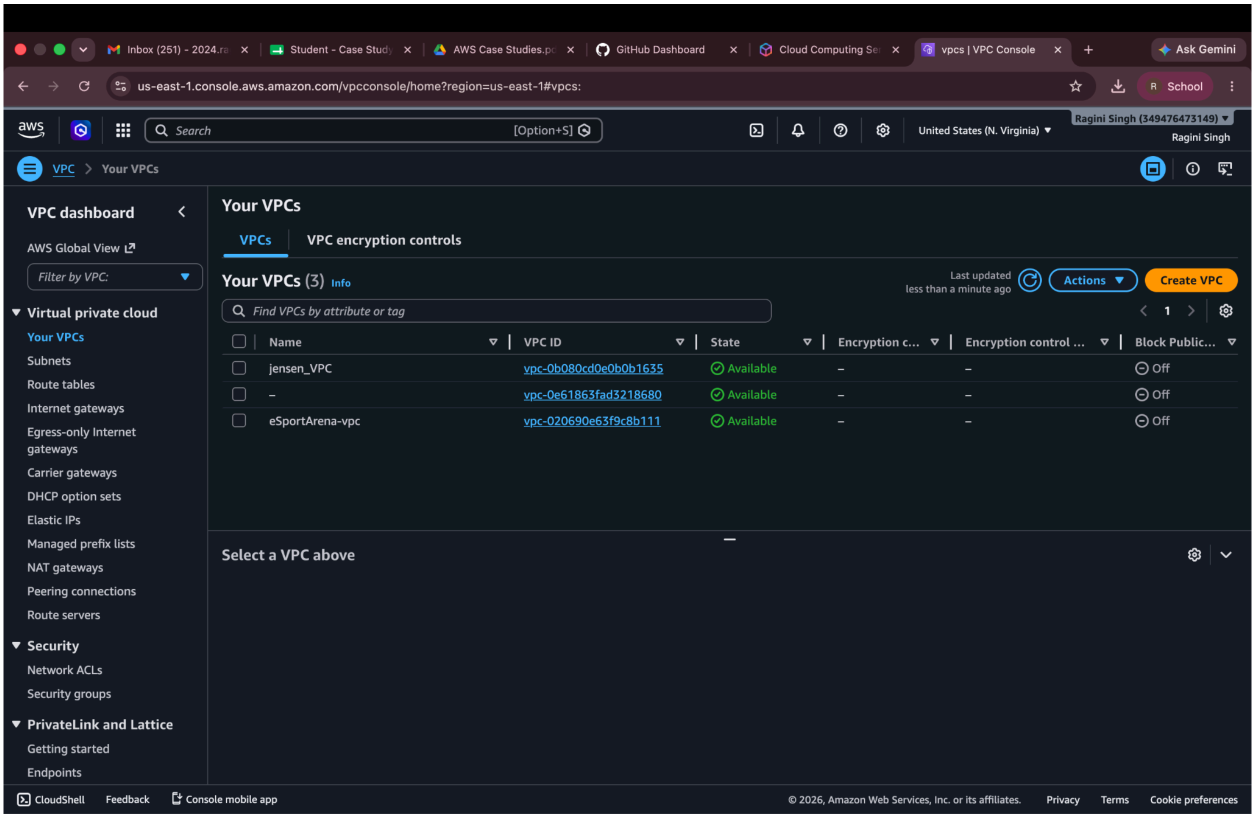
1. Aws_login_page.png



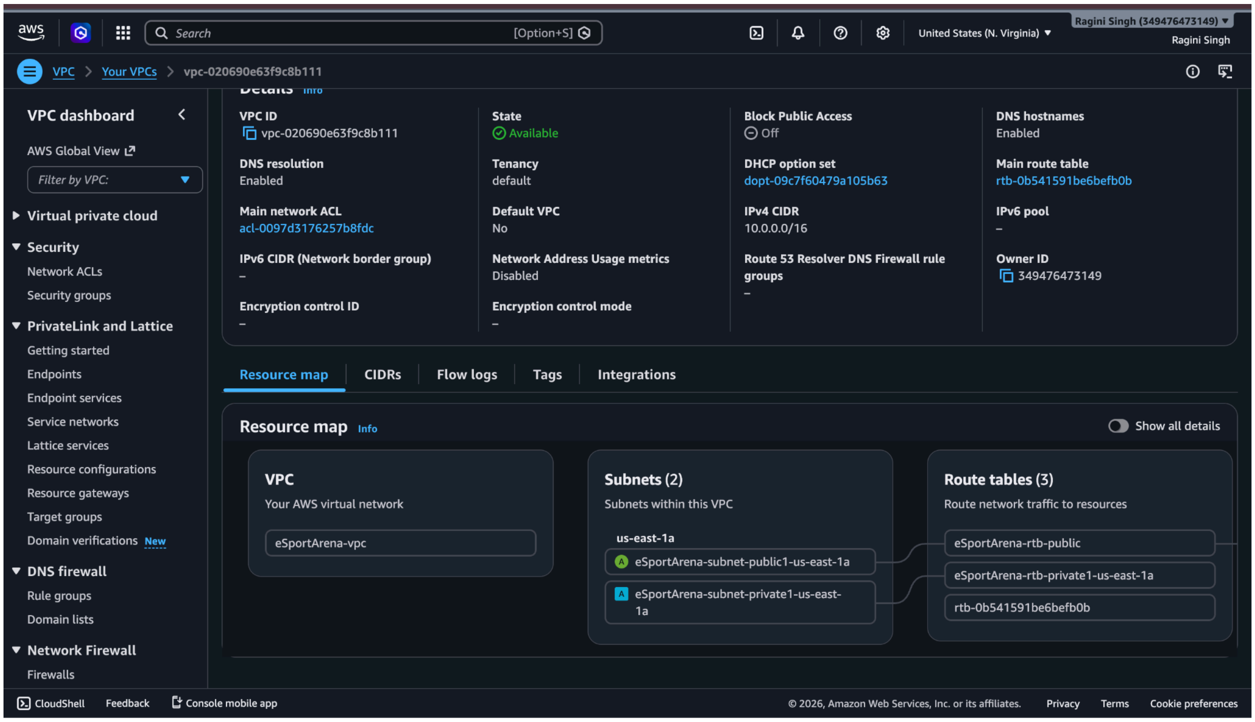
2. SS-02-VPC-Created.png



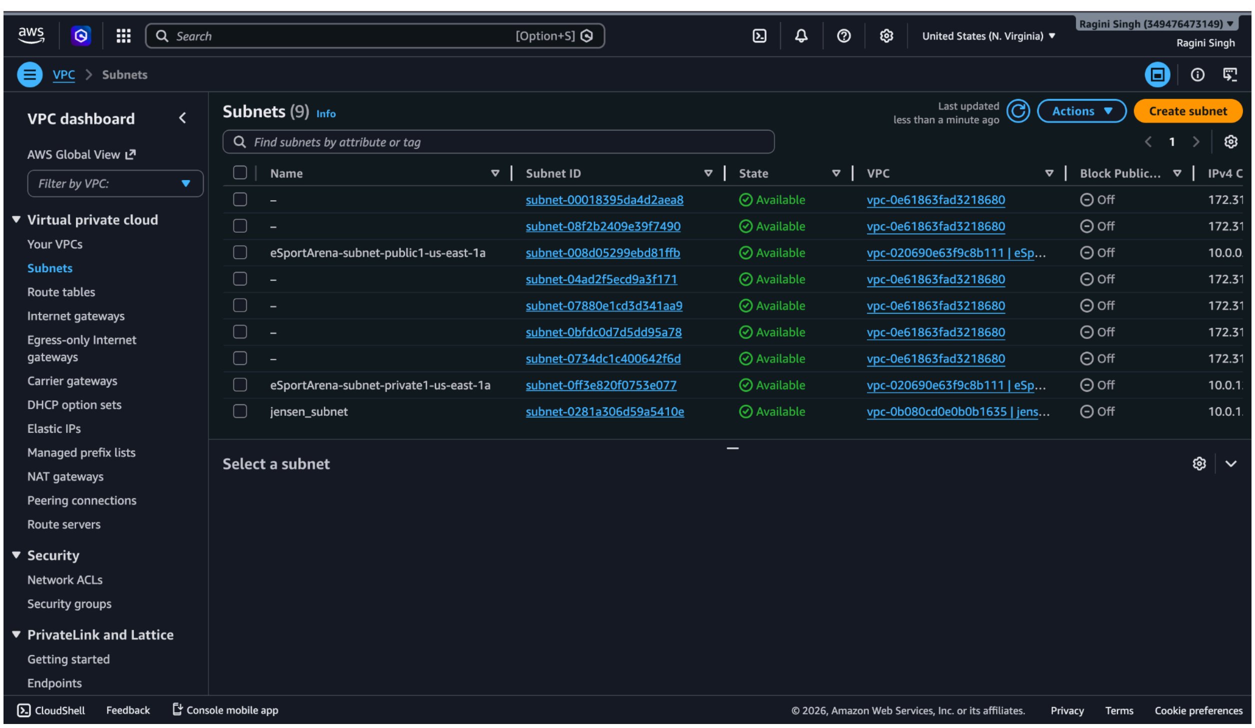
3. SS-02-VPC-List.png



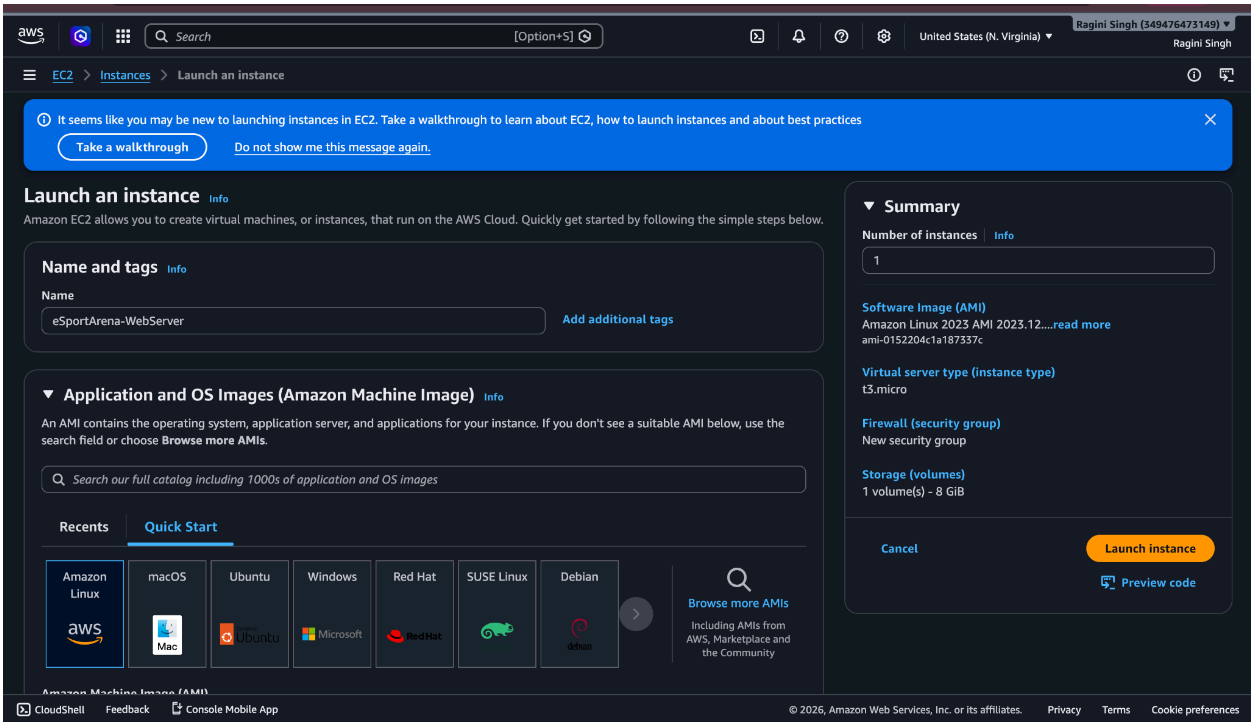
4. SS-03-Resource-Map.png



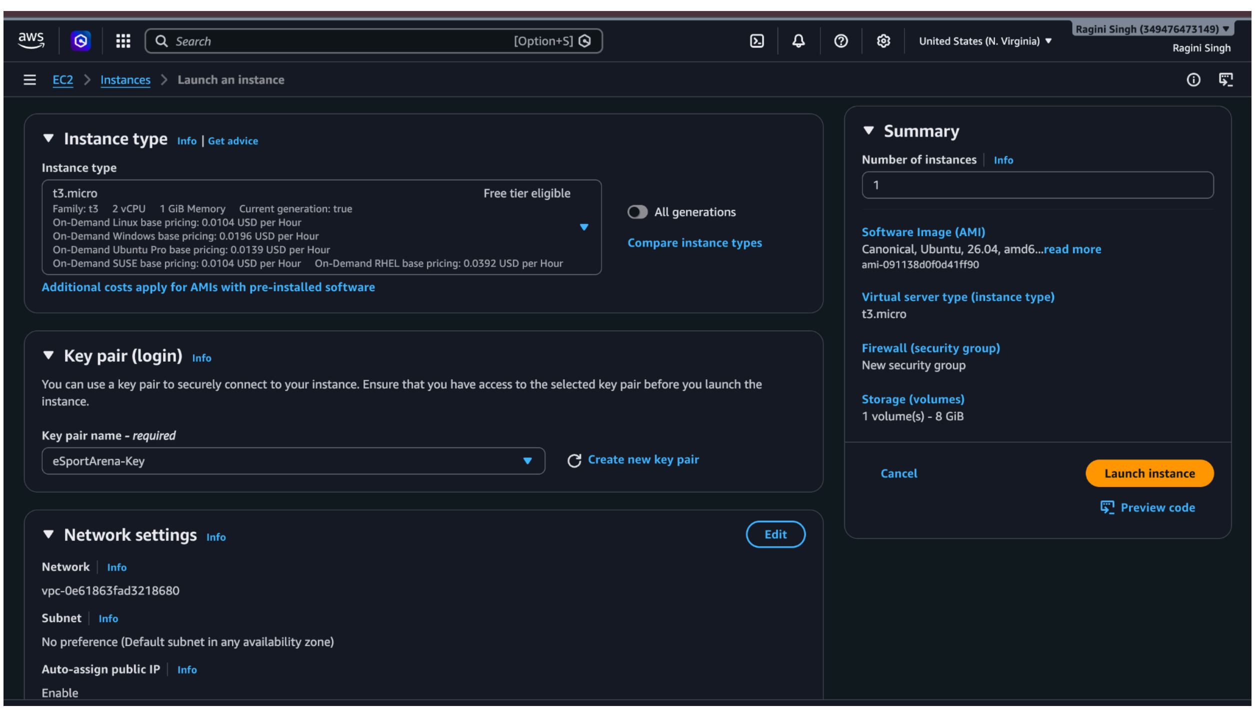
5. SS-03-Subnets.png



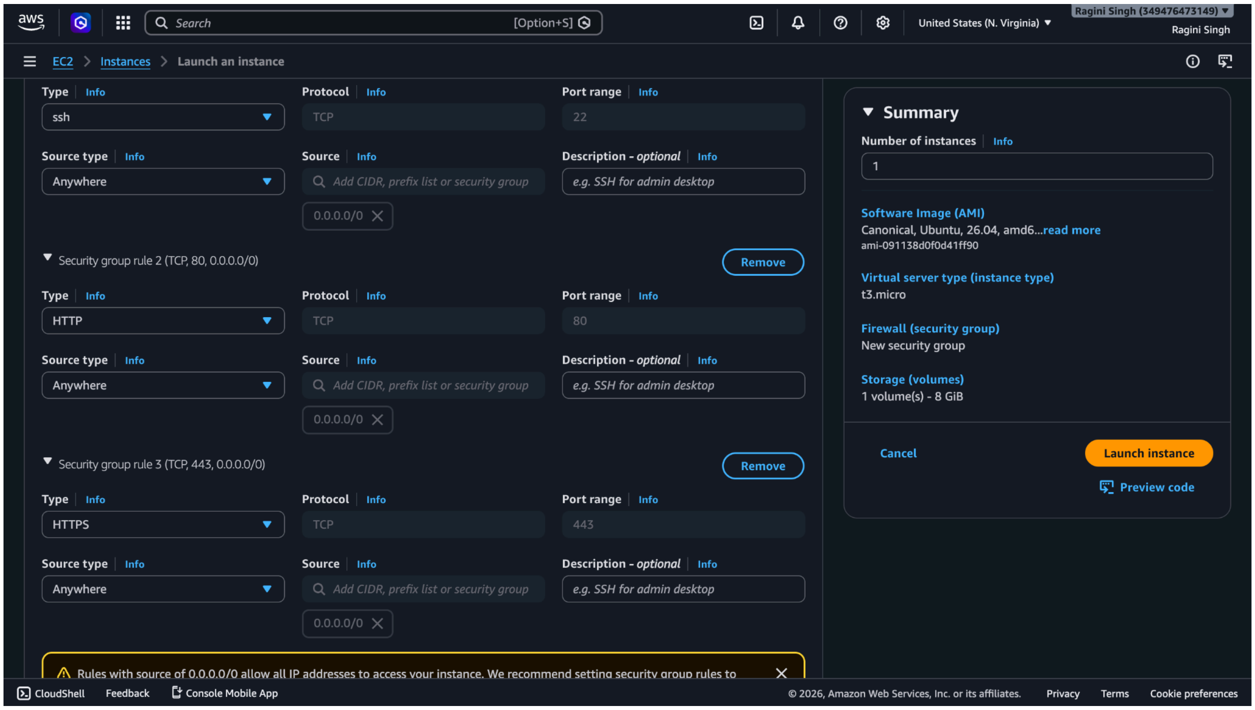
6. SS-04-EC2-Configuration.png



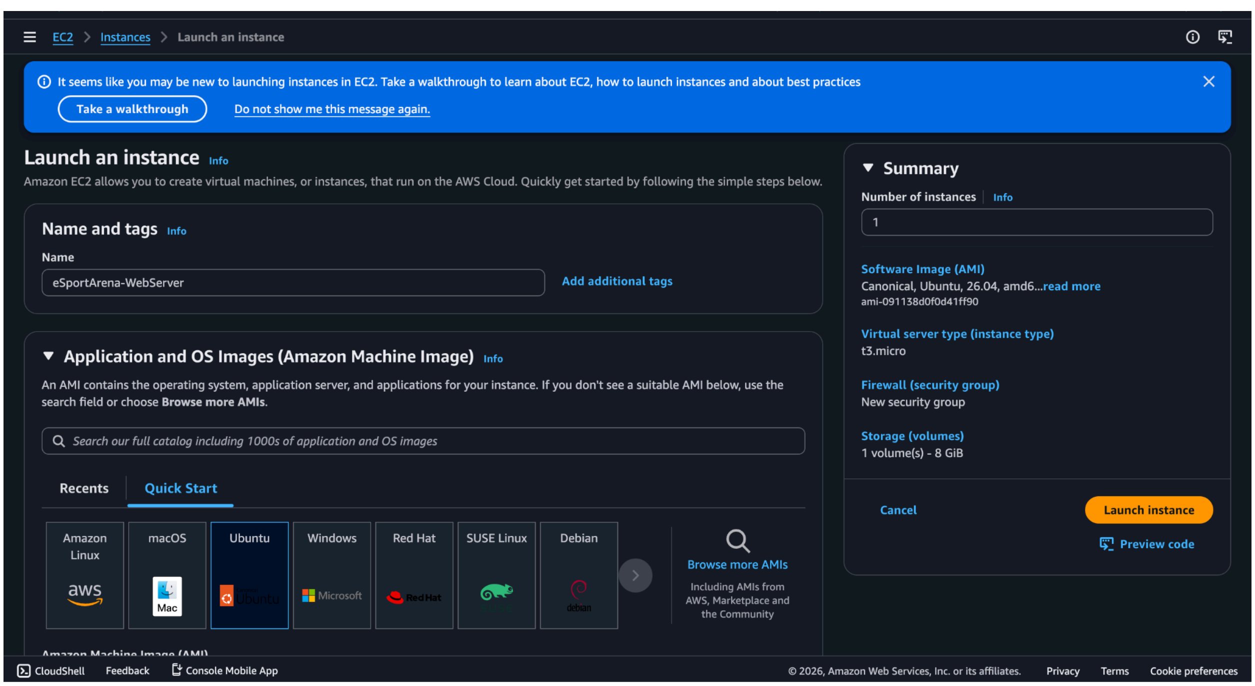
7. SS-05-KeyPair-Created.png



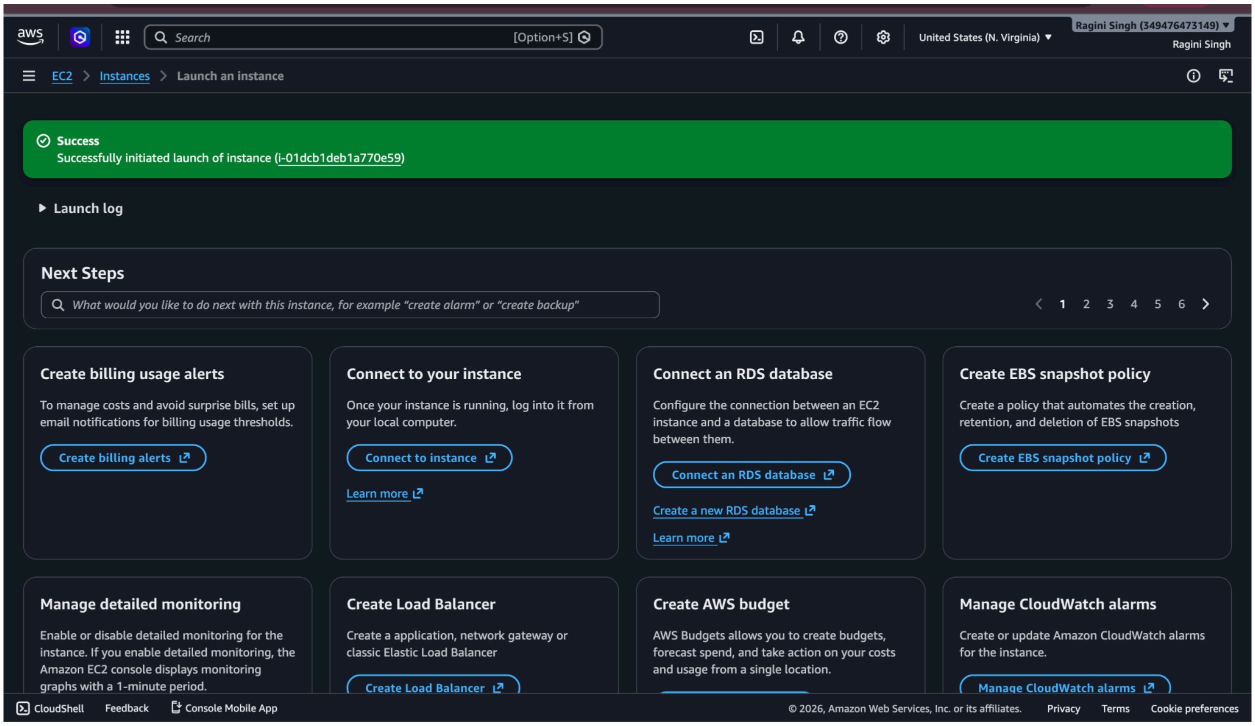
8. SS-07-Security-Group-Rules.png



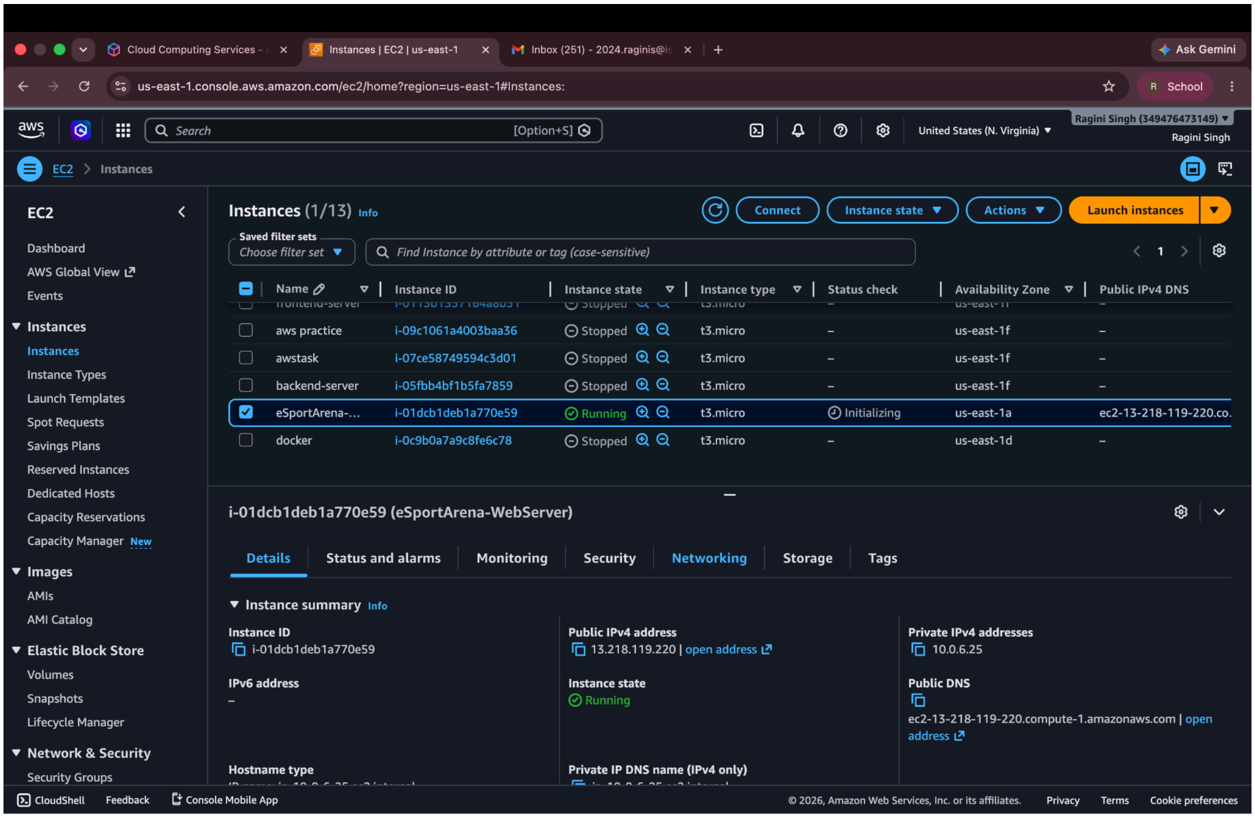
9. SS-08-EC2-Launch-Configuration.png



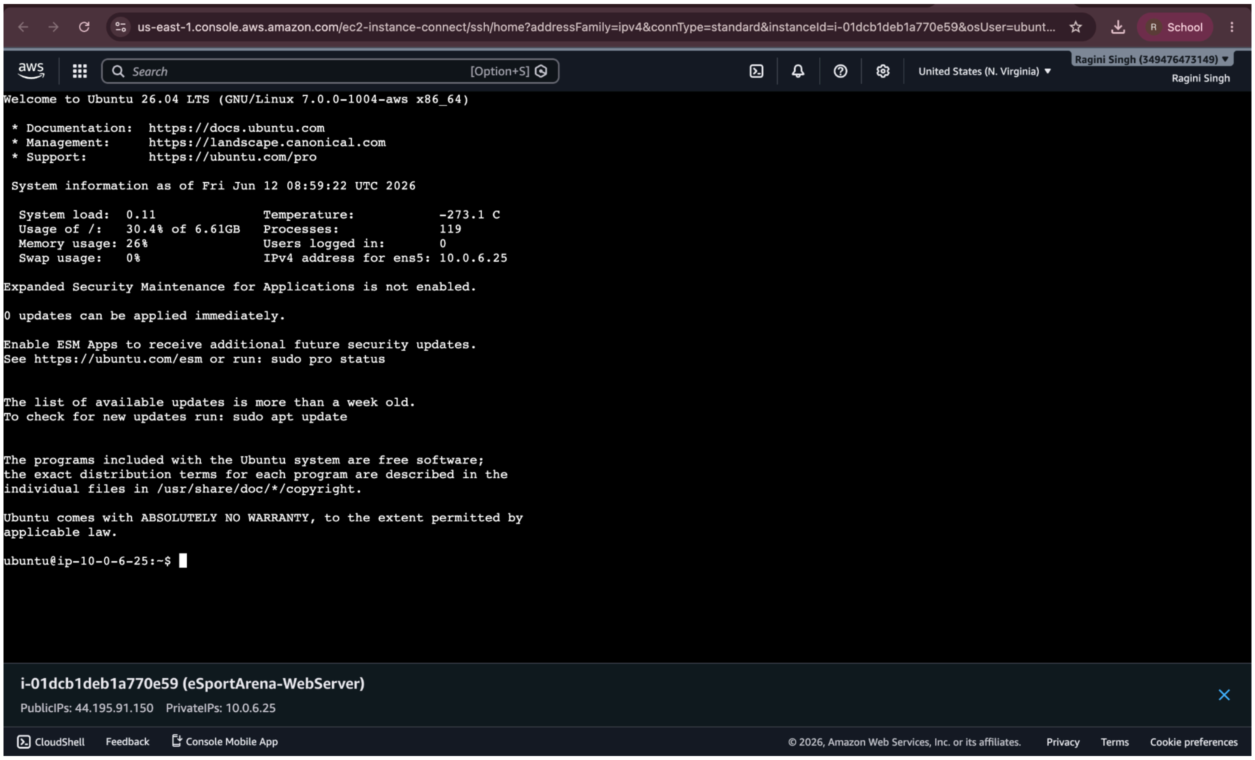
10. SS-09-EC2-Running-Instance.png



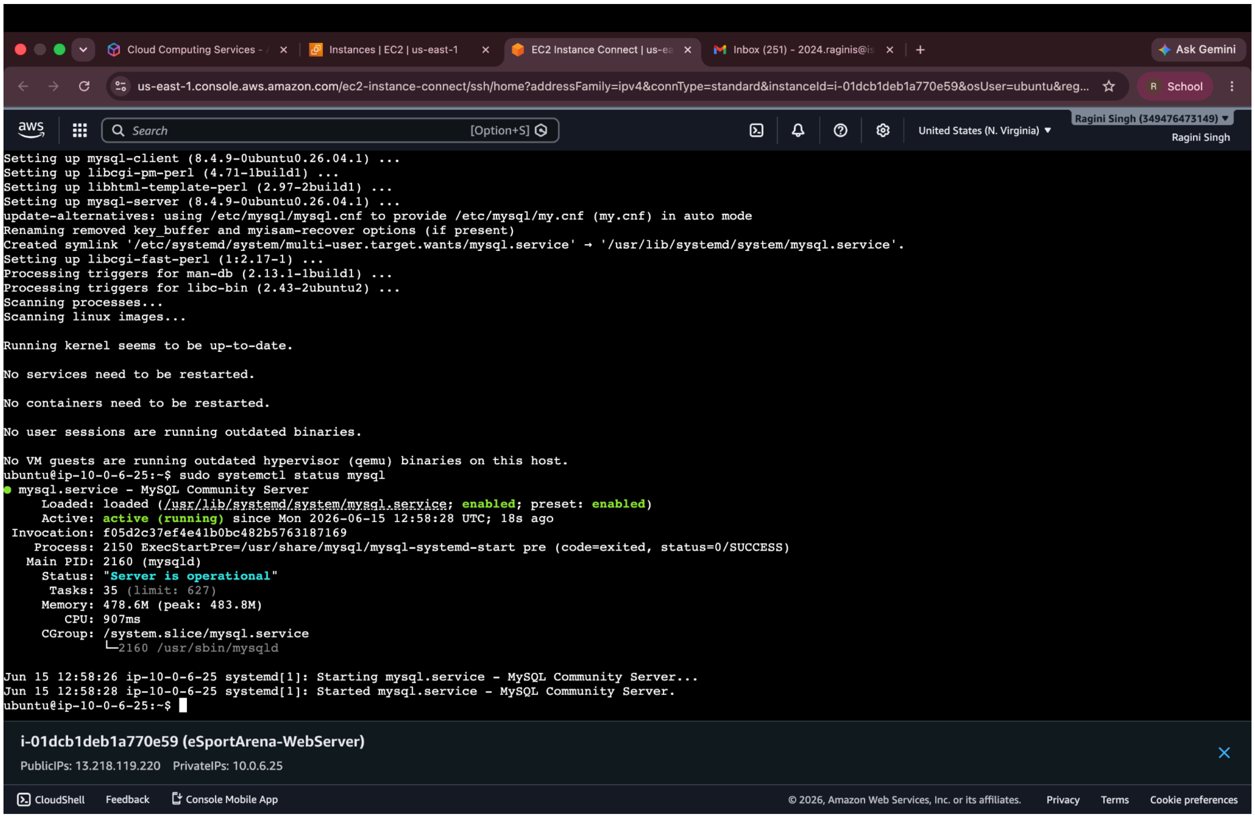
11. SS-09-EC2-Started-Again.png



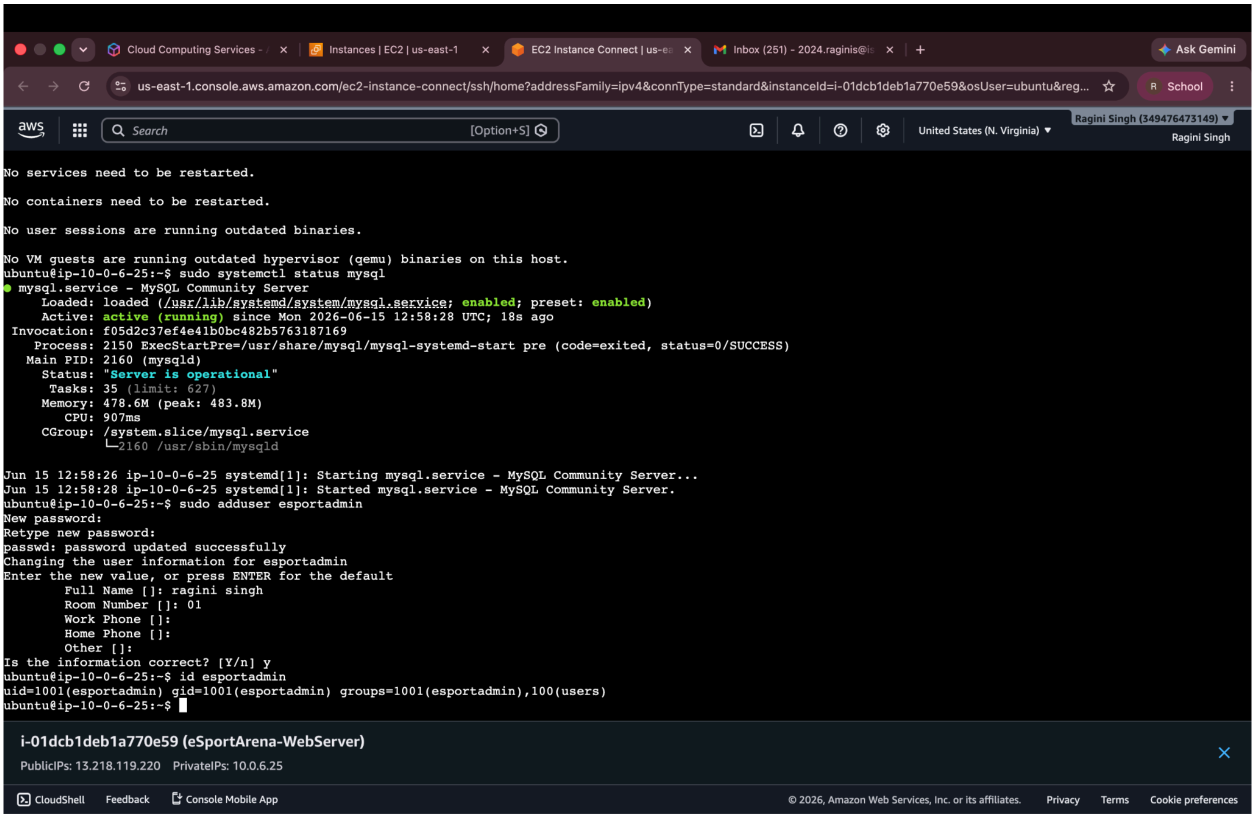
12. SS-10-EC2-Terminal-Connected.png


us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-01dcb1deb1a770e59&osUser=ubunt...
School
Search
[Option+S]
United States (N. Virginia)
Ragini Singh (349476473149)
Ragini Singh
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1004-aws x86_64)
* Documentation: https://docs.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
System information as of Fri Jun 12 08:59:22 UTC 2026
System load: 0.11 Temperature: -273.1 C
Usage of /: 30.4% of 6.61GB Processes: 119
Memory usage: 26% Users logged in: 0
Swap usage: 0% IPv4 address for ens5: 10.0.6.25
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
The list of available updates is more than a week old.
To check for new updates run: sudo apt update
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
ubuntu@ip-10-0-6-25:~$
i-01dcb1deb1a770e59 (eSportArena-WebServer)
PublicIPs: 44.195.91.150 PrivateIPs: 10.0.6.25
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

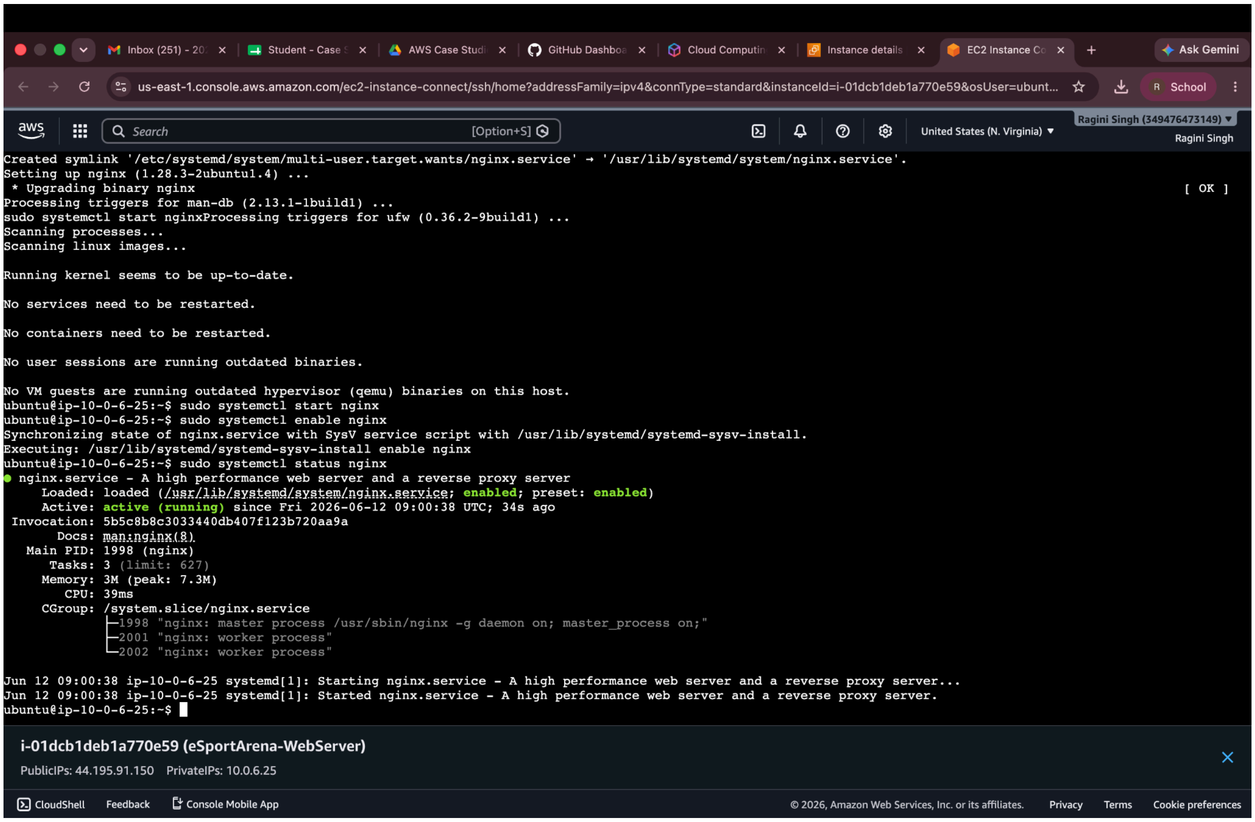
13. SS-10-MySQL-Running.png



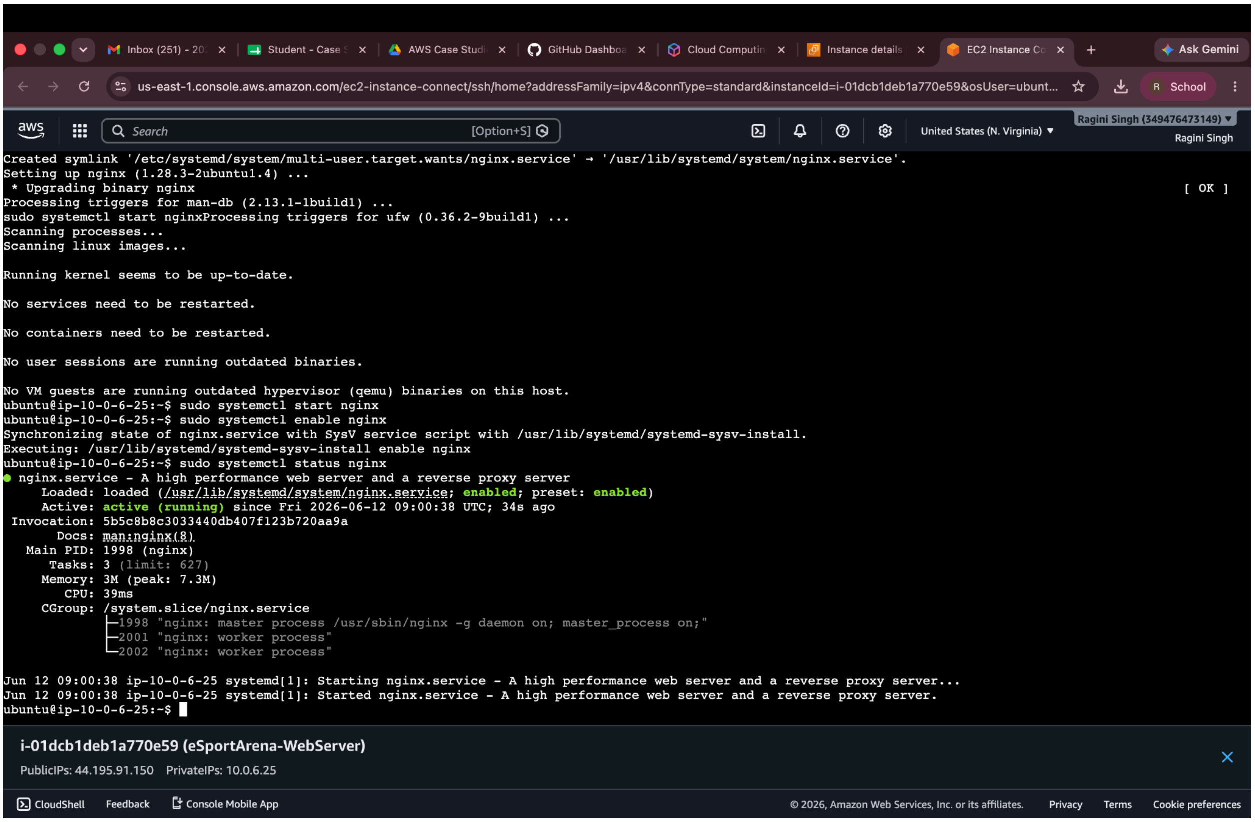
14. SS-11-Linux-User-Created.png



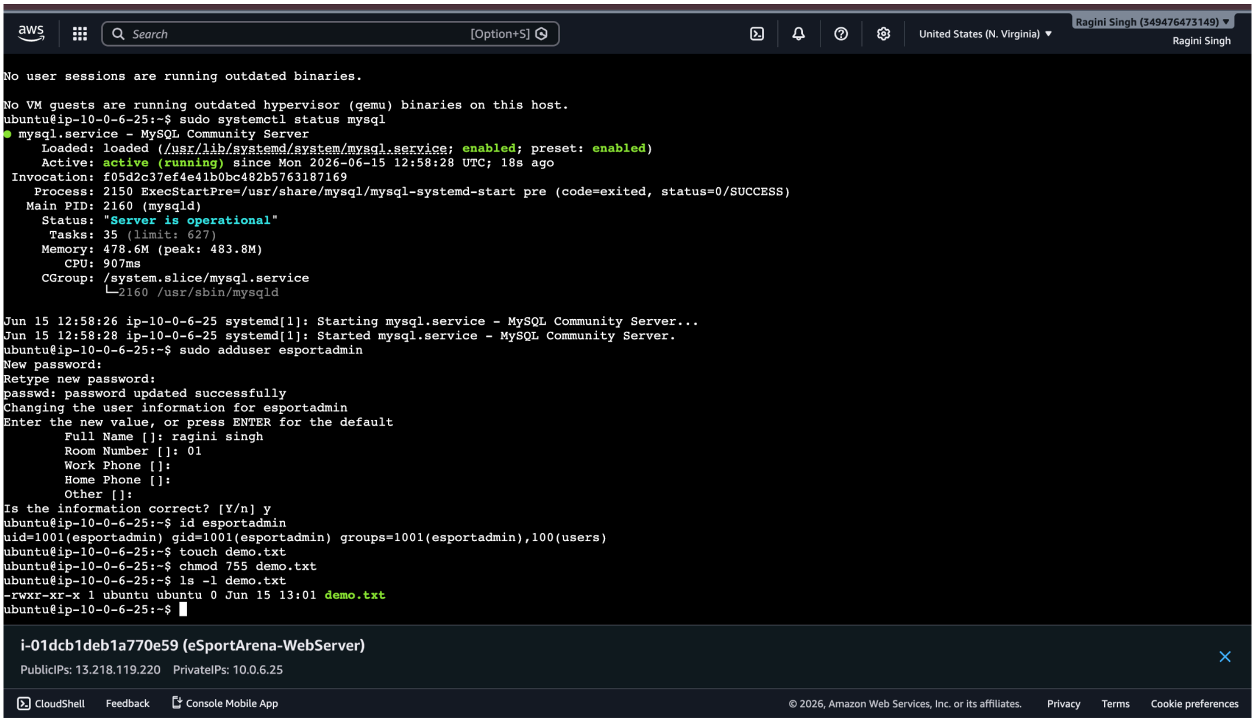
15. SS-11-Nginx-Service-Running.png



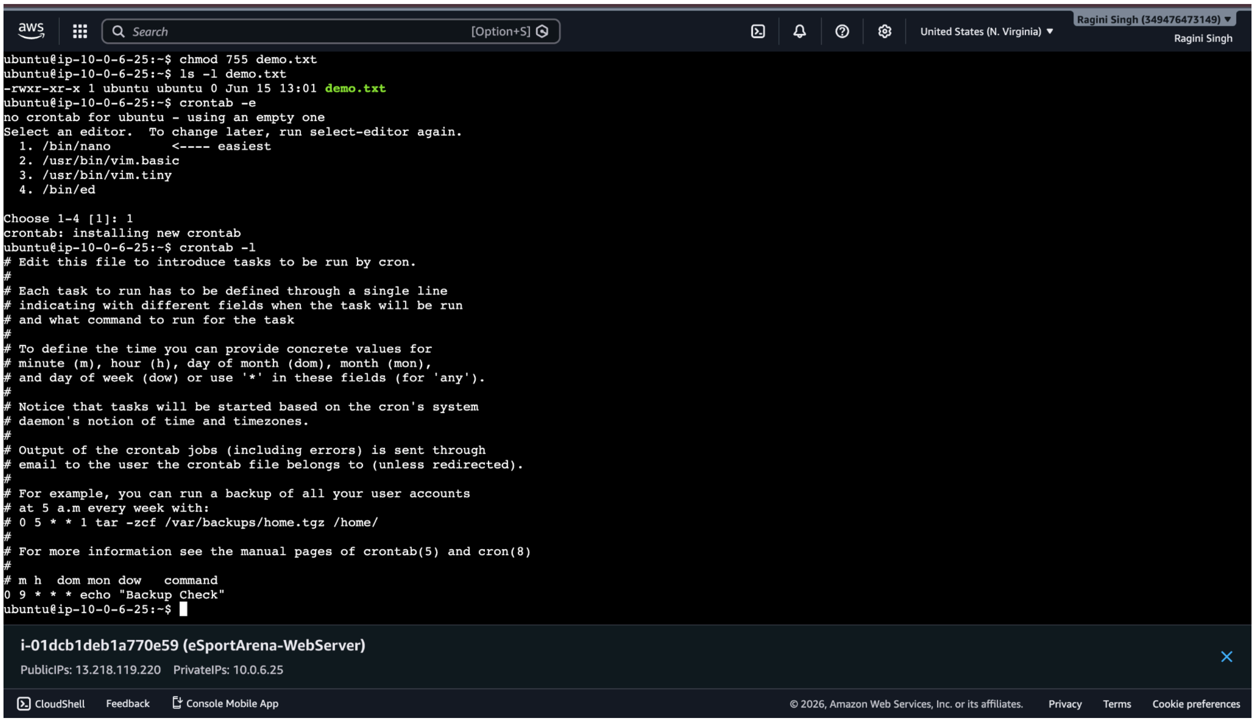
15. SS-11-Nginx-Service-Running.png



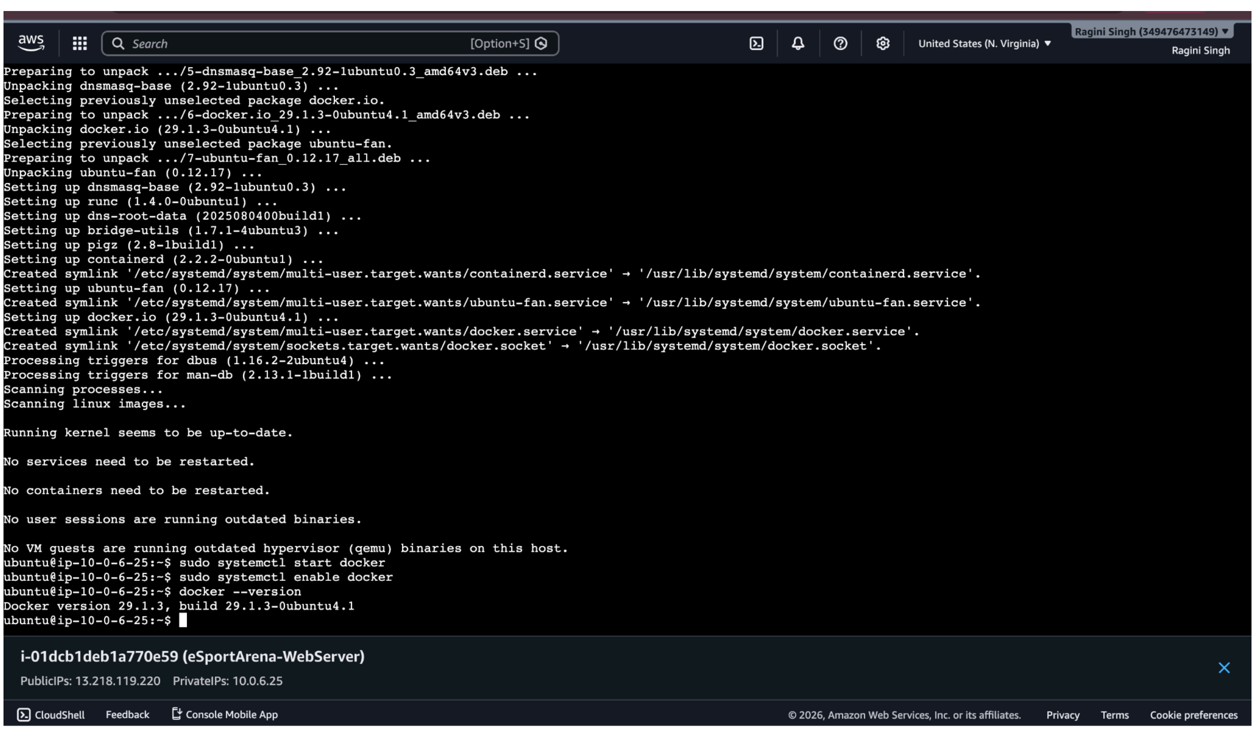
16. SS-12-Permissions.png



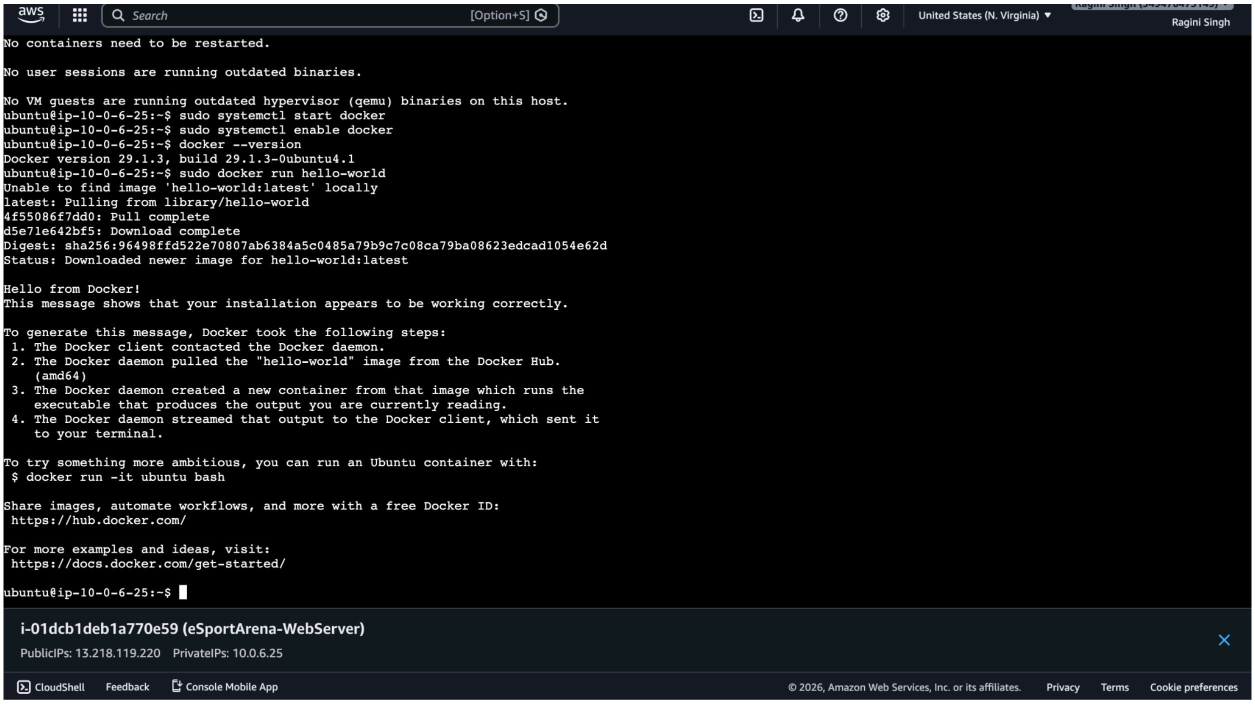
17. SS-13-CronJob.png



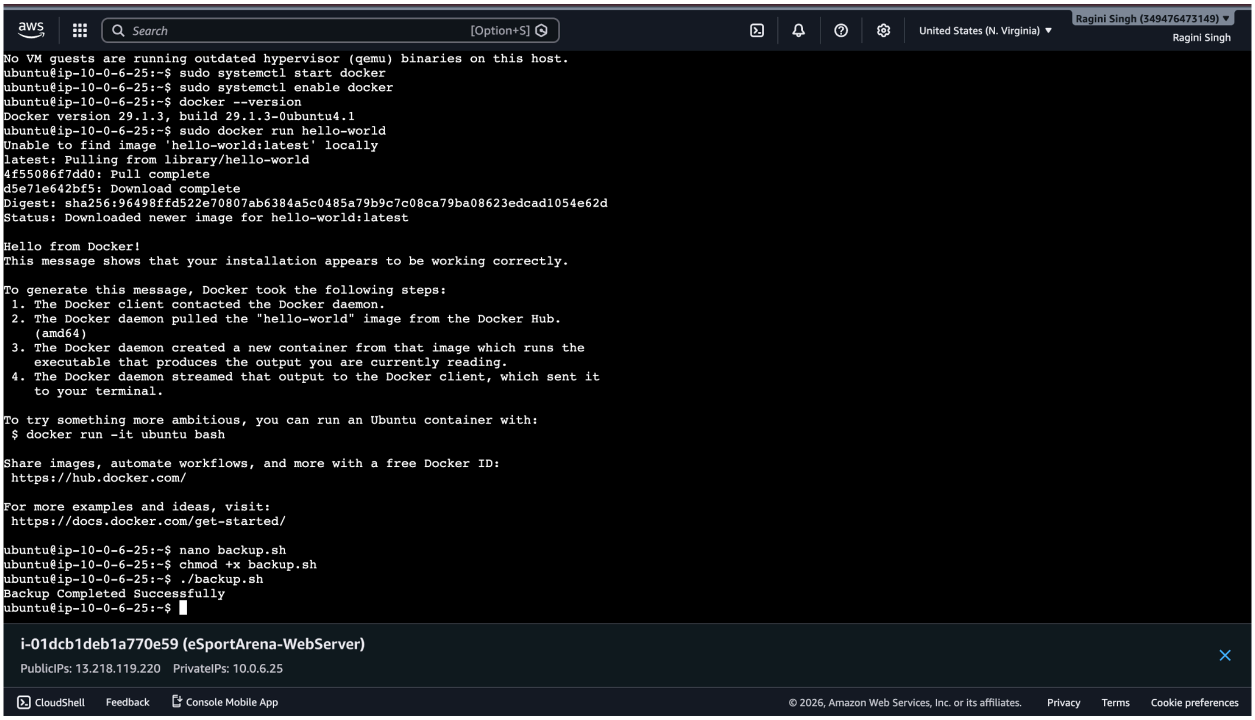
18. SS-14-Docker-Installed.png



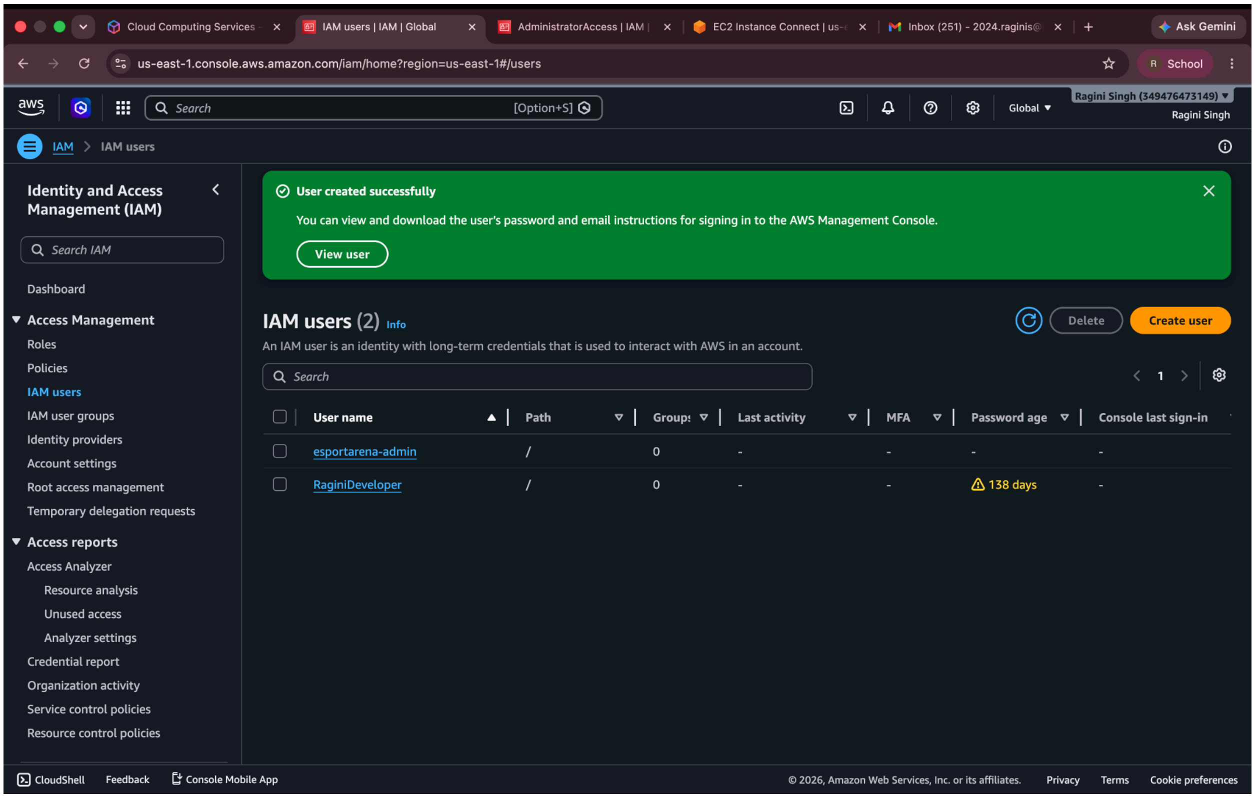
19. SS-15-Docker-Container.png



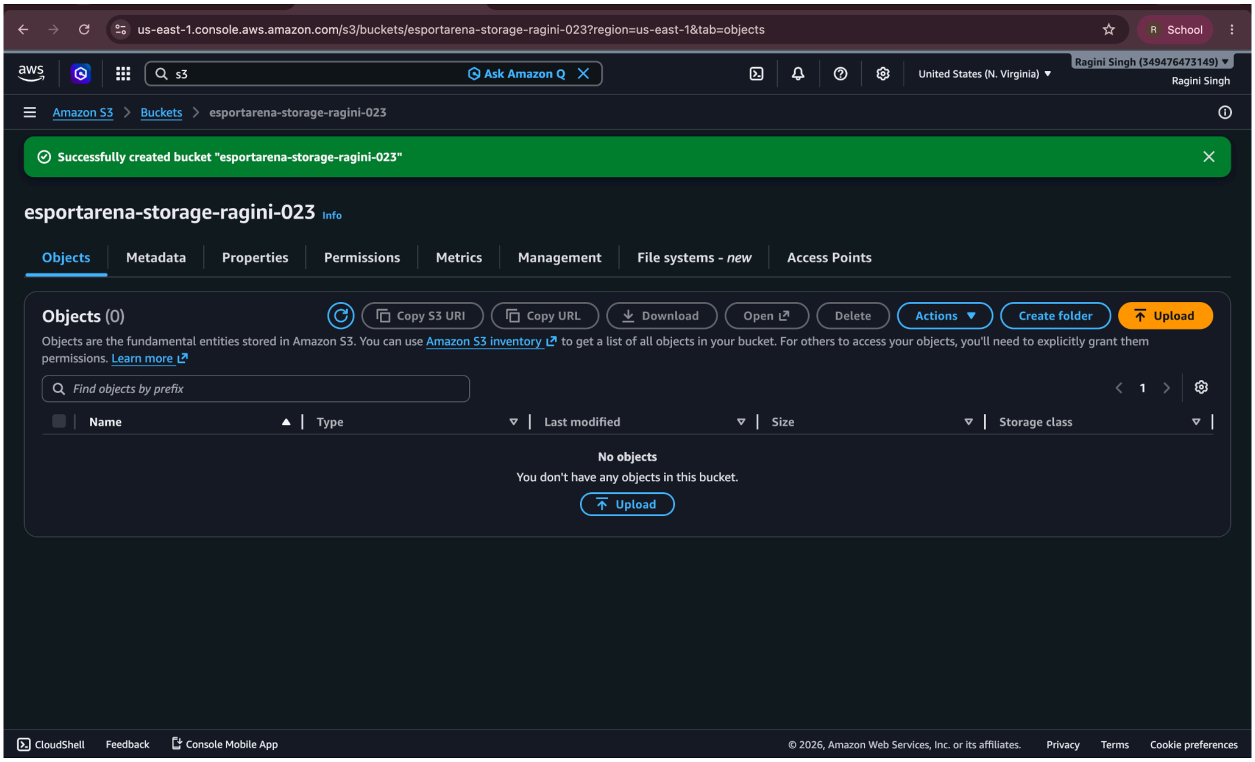
20. SS-16-Shell-Script.png



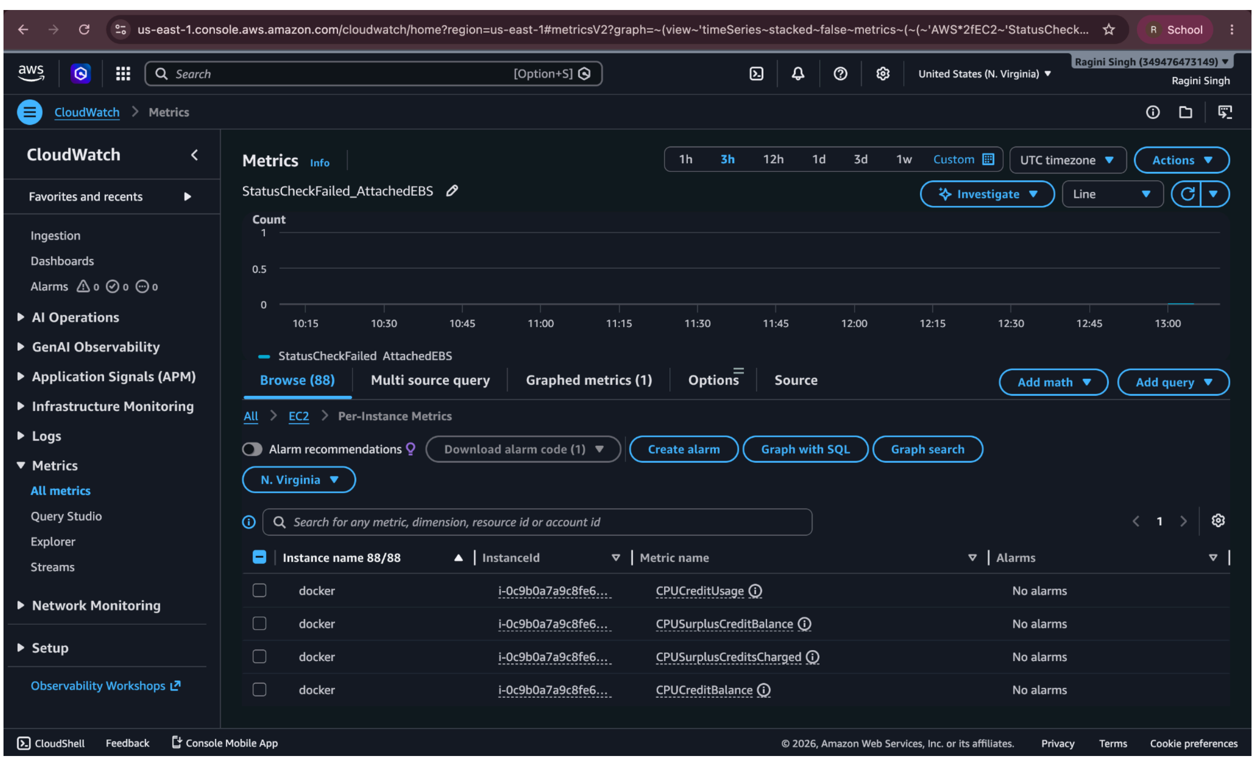
21. SS-17-IAM-User.png



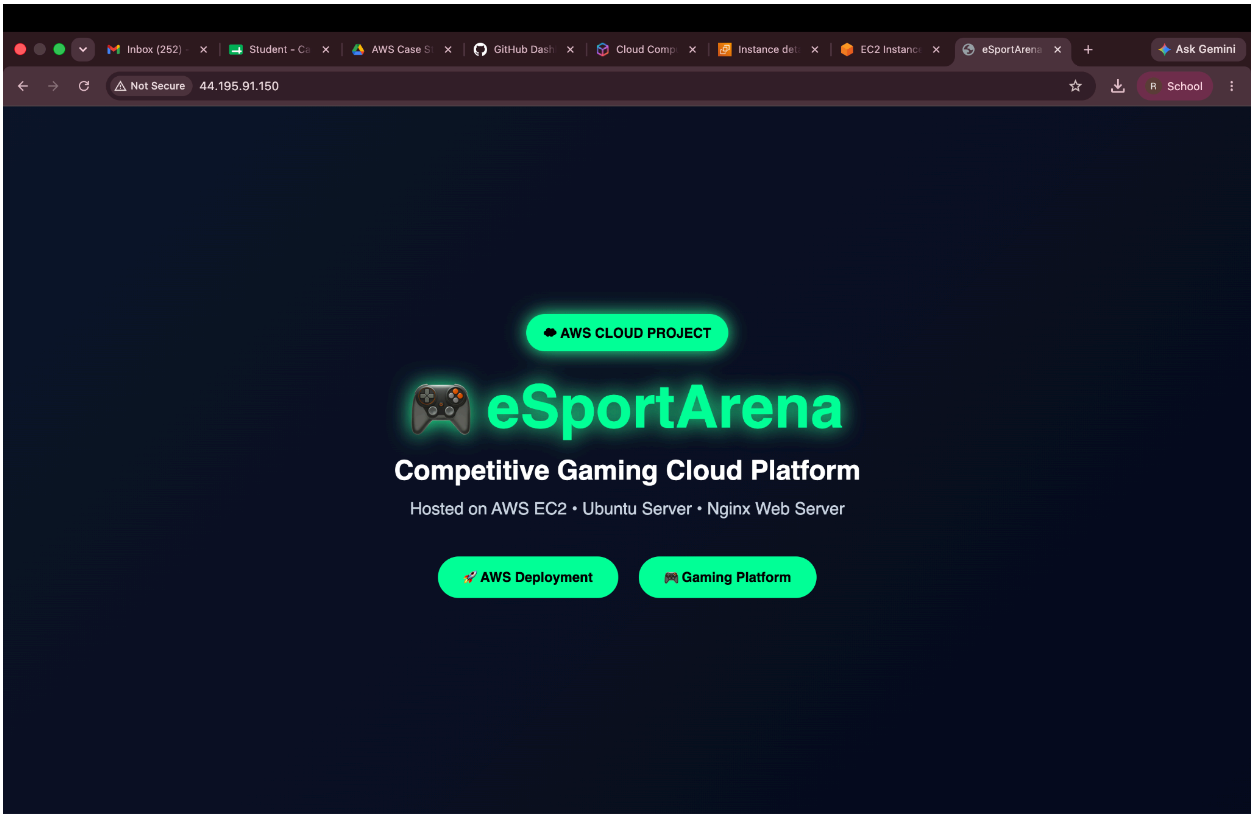
22. SS-18-S3-Bucket.png



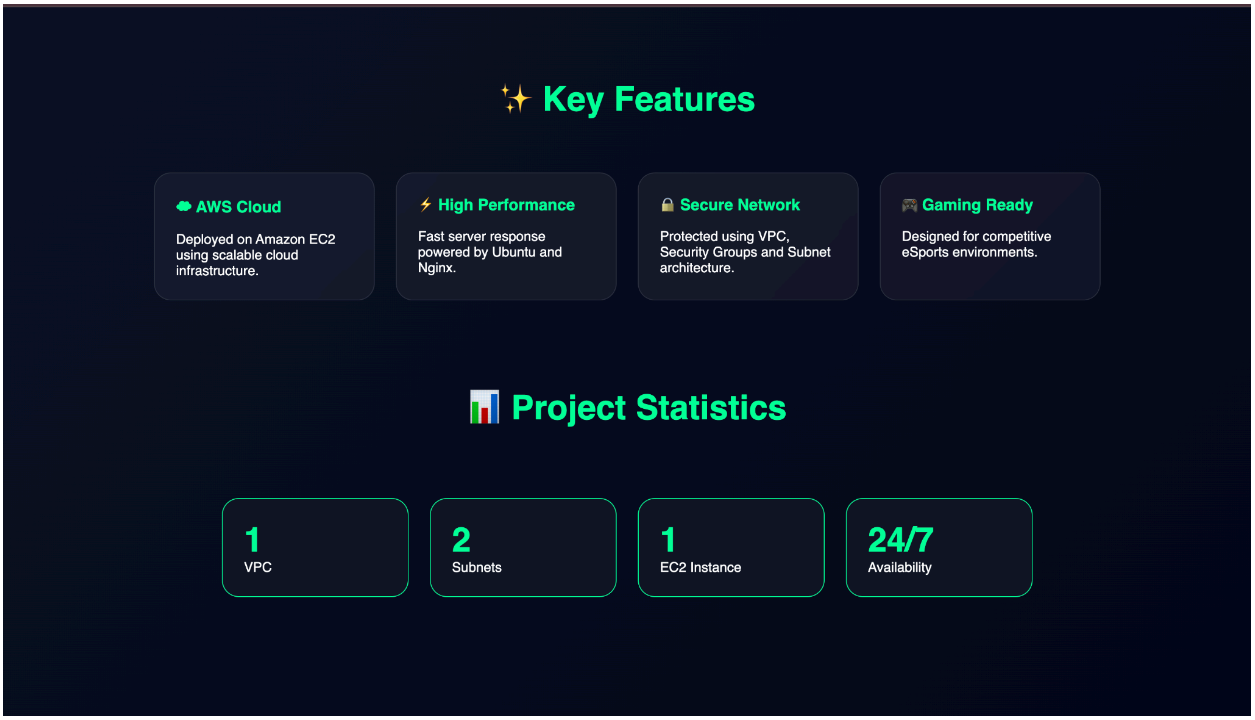
23. SS-19-CloudWatch-Metrics.png



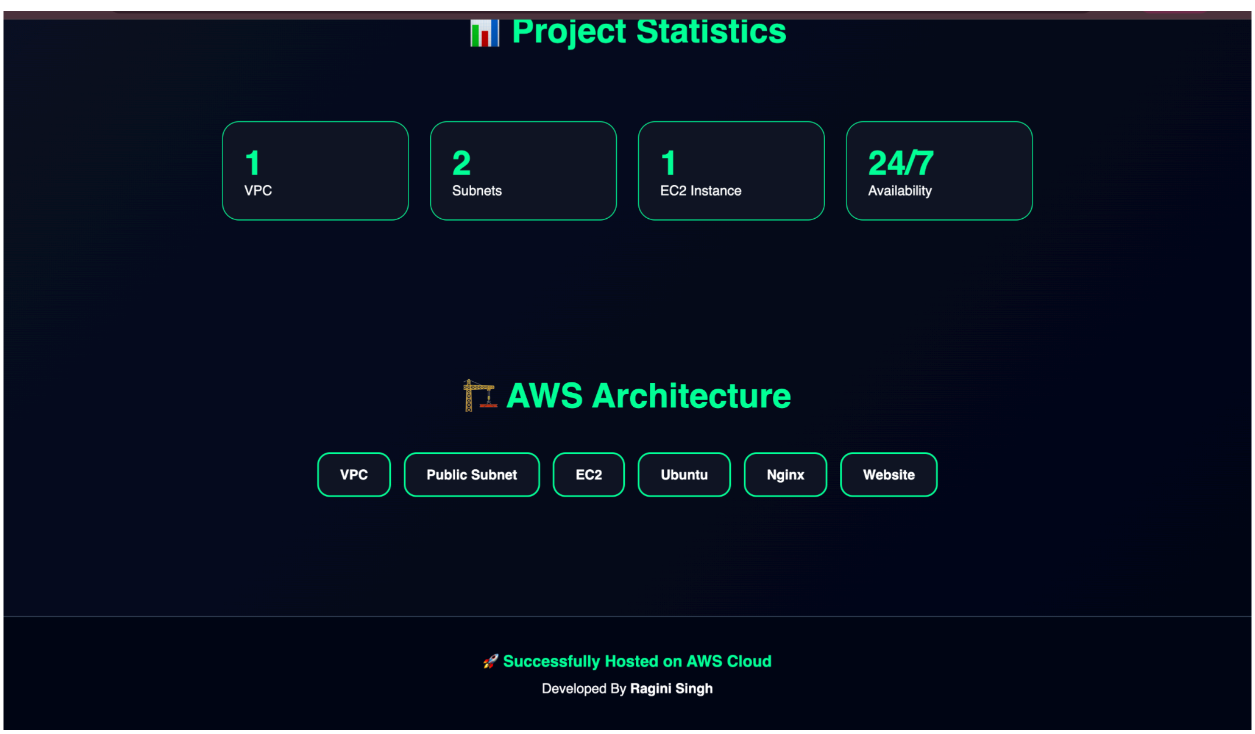
24. Website Home Page.png



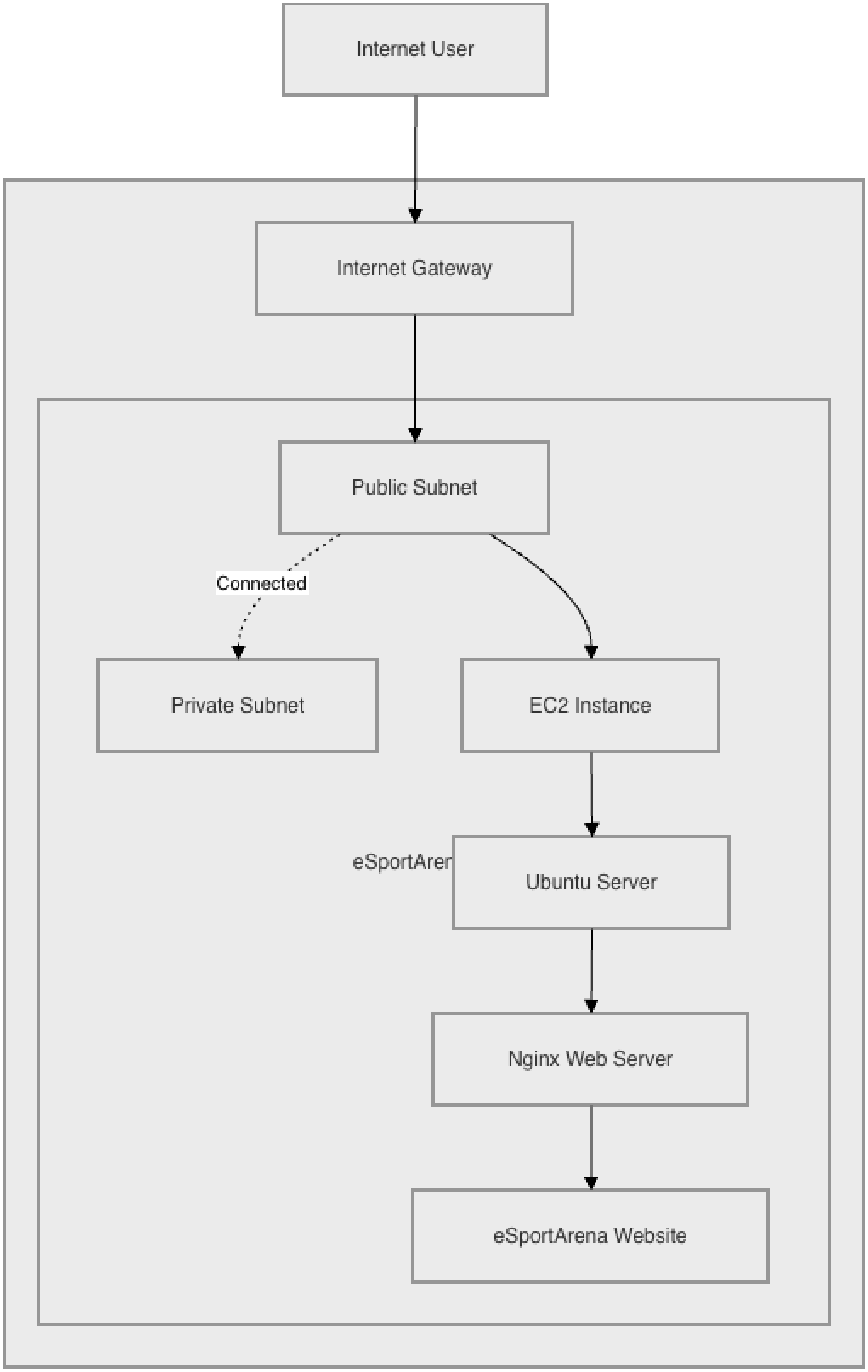
25. Website Features Section.png



26. Website Architecture Section.png



27. AWS_Architecture.drawio.png



7. Results

The eSportArena web application was successfully deployed on AWS EC2 using Ubuntu and Nginx. The website became accessible through the assigned public IP address. The project demonstrates practical implementation of cloud computing concepts including networking, virtualization, security, and web hosting.

8. Future Scope

- Add Load Balancer
- Implement Auto Scaling
- Configure Route 53 Domain
- Deploy HTTPS using SSL Certificates
- Integrate CloudWatch Monitoring
- Containerize Application using Docker
- Deploy using Kubernetes

9. Conclusion

The project successfully demonstrates the deployment of a cloud-hosted gaming platform infrastructure using AWS services. It provides hands-on experience with EC2, VPC, Security Groups, Ubuntu, and Nginx while showcasing the benefits of scalability, availability, and security offered by cloud computing.

THANKYOUUU!!!!!